

Research Notes on Text summarization

Text summarization

>> Section 1

Multi-document summarization is an automatic procedure aimed at extraction of information from multiple texts written about the same topic. Resulting summary report allows individual users, such as professional information consumers, to quickly familiarize themselves with information contained in a large cluster of documents. In such a way, multi-document summarization systems are complementing the news aggregators performing the next step down the road of coping with information overload. Multi-document summarization may also be done in response to a question. Multi-document summarization creates information reports that are both concise and comprehensive. With different opinions being put together and outlined, every topic is described from multiple perspectives within a single document. While the goal of a brief summary is to simplify information search and cut the time by pointing to the most relevant source documents, comprehensive multi-document summary should itself contain the required information, hence limiting the need for accessing original files to cases when refinement is required. Automatic summaries present information extracted from multiple sources algorithmically, without any editorial touch or subjective human intervention, thus making it completely unbiased.

>> Section 2

Submodular functions as generic tools for summarization

The idea of a submodular set function has recently emerged as a powerful modeling tool for various summarization problems. Submodular functions naturally model notions of coverage, information, representation and diversity. Moreover, several important combinatorial optimization problems occur as special instances of submodular optimization. For example, the set cover problem is a special case of submodular optimization, since the set cover function is submodular. The set cover function attempts to find a subset of objects which cover a given set of concepts. For example, in document summarization, one would like the summary to cover all important and relevant concepts in the document. This is an instance of set cover. Similarly, the facility location problem is a special case of submodular functions. The Facility Location function also naturally models coverage and diversity. Another example of a submodular optimization problem is using a determinantal point process to model diversity. Similarly, the Maximum-Marginal-Relevance procedure can also be seen as an instance of submodular optimization. All these important models encouraging coverage, diversity and information are all submodular. Moreover, submodular functions can be efficiently combined, and the resulting

function is still submodular. Hence, one could combine one submodular function which models diversity, another one which models coverage and use human supervision to learn a right model of a submodular function for the problem. While submodular functions are fitting problems for summarization, they also admit very efficient algorithms for optimization. For example, a simple greedy algorithm admits a constant factor guarantee. Moreover, the greedy algorithm is extremely simple to implement and can scale to large datasets, which is very important for summarization problems. Submodular functions have achieved state-of-the-art for almost all summarization problems. For example, work by Lin and Bilmes, 2012 shows that submodular functions achieve the best results to date on DUC-04, DUC-05, DUC-06 and DUC-07 systems for document summarization. Similarly, work by Lin and Bilmes, 2011, shows that many existing systems for automatic summarization are instances of submodular functions. This was a breakthrough result establishing submodular functions as the right models for summarization problems. Submodular Functions have also been used for other summarization tasks. Tschiatschek et al., 2014 show that mixtures of submodular functions achieve state-of-the-art results for image collection summarization. Similarly, Bairi et al., 2015 show the utility of submodular functions for summarizing multi-document topic hierarchies. Submodular Functions have also successfully been used for summarizing machine learning datasets.

>> Section 3

Automatic summarization is the process of shortening a set of data computationally, to create a subset (a summary) that represents the most important or relevant information within the original content. Artificial intelligence (AI) algorithms are commonly developed and employed to achieve this, specialized for different types of data. Text summarization is usually implemented by natural language processing methods, designed to locate the most informative sentences in a given document. On the other hand, visual content can be summarized using computer vision algorithms. Image summarization is the subject of ongoing research; existing approaches typically attempt to display the most representative images from a given image collection, or generate a video that only includes the most important content from the entire collection. Video summarization algorithms identify and extract from the original video content the most important frames (key-frames), and/or the most important video segments (key-shots), normally in a temporally ordered fashion. Video summaries simply retain a carefully selected subset of the original video frames and, therefore, are not identical to the output of video synopsis algorithms, where new

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Text summarization

video frames are being synthesized based on the original video content.

>> Section 4

History The first publication in the area dates back to 1957 (Hans Peter Luhn), starting with a statistical technique. Research increased significantly in 2015. Term frequency-inverse document frequency had been used by 2016. Pattern-based summarization was the most powerful option for multi-document summarization found by 2016. In the following year it was surpassed by latent semantic analysis (LSA) combined with non-negative matrix factorization (NMF). Although they did not replace other approaches and are often combined with them, by 2019 machine learning methods dominated the extractive summarization of single documents, which was considered to be nearing maturity. By 2020, the field was still very active and research is shifting towards abstractive summation and real-time summarization.

>> Section 5

Recent approaches Recently the rise of transformer models replacing more traditional RNN (LSTM) have provided a flexibility in the mapping of text sequences to text sequences of a different type, which is well suited to automatic summarization. This includes models such as T5 and Pegasus.

>> Section 6

Abstractive-based summarization Abstractive summarization methods generate new text that did not exist in the original text. This has been applied mainly for text. Abstractive methods build an internal semantic representation of the original content (often called a language model), and then use this representation to create a summary that is closer to what a human might express. Abstraction may transform the extracted content by paraphrasing sections of the source document, to condense a text more strongly than extraction. Such transformation, however, is computationally much more challenging than extraction, involving both natural language processing and often a deep understanding of the domain of the original text in cases where the original document relates to a special field of knowledge. "Paraphrasing" is even more difficult to apply to images and videos, which is why most summarization systems are extractive.